Algebra 1 | Name_<u>Answer Key</u>_________
Unit 13 | Date____________________
Lesson 8 Homework | Period__________________

Suni is biking 20 miles. To fuel her bike ride, she will need to consume 738 calories. She plans to bring sports drink and granola bars. Sports drink bottles are 158 calories and granola bars are 132 calories.

1. $158s + 132g$ is a(n)
   - ○ function.
   - ○ equation.
   - ★ expression.

2. The values 158 and 132 are
   - ★ coefficients.
   - ○ terms.
   - ○ constants.

3. The monomials $158s$ and $132g$ are
   - ○ coefficients.
   - ★ terms.
   - ○ constants.

4. $158s + 132g = 738$ is a(n)
   - ○ function.
   - ★ equation.
   - ○ expression.

5. Determine the amount of granola bars Suni will need if she has three bottles of sports drink.

   $158(3) + 132g = 738$

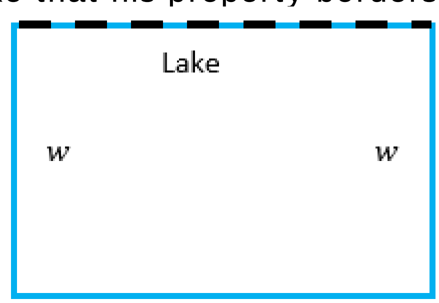
   $474 + 132g = 738$

   $-474 \qquad\qquad -474$

   $\frac{132g}{132} = \frac{264}{132}$

   2 granola bars

A rancher has 3000 feet of fencing to enclose his property, which borders a lake. The diagram shows the width, w, of the property and the lake that his property borders. The fencing is not used along the lake.



Explain what each quantity represents in this context.

6. $3000 - 2w$ = amount of fence left for last side.

   3000 = total fence to be used

   $2w$ = two sides that will use fence, where w is the length of those sides.

7. $w(3000 - 2w)$

   Width times amount of fence used for the last side.

Luis started a retirement savings fund when he began his first job in 2015. The amount of money in the retirement savings fund since 2015 can be represented by the equation $y = 15{,}000(1 + 0.025)^x$, where y is the amount in the retirement savings fund and x is the number of years since 2015.

8. The equation $y = 15{,}000(1 + 0.025)^x$ represents

★ exponential growth.
○ exponential decay.

9. The rate of

○ decay
★ growth

is 2.5 %.

10. What does the coefficient $15{,}000$ represent?

Initial amount in savings account.